

UNC801: DATA SCIENCE FOR ENGINEERS				
	L	T	P	Cr
	3	0	0	3.0
Course Objective: The objective of the Data Science for Engineers course is to equip engineering students with the foundational knowledge and practical skills needed to analyze, interpret, and leverage data for informed decision-making and problem-solving in engineering contexts..				
Syllabus Introduction to R: Variables and Datatypes in R, Data frames, Recasting and Joining of Dataframes, Arithmetic, Logical and Matrix Operations in R, Advanced Programming in R : Functions, Control Structures Data Visualization in R Basic Graphics Linear Algebra for Data science Solving Linear Equations, Linear Algebra - Distance, Hyperplanes and Halfspaces, Eigenvalues, Eigenvectors Statistical Modelling Random Variables and Probability Mass/Density Functions Sample Statistics, Hypotheses Testing Optimization for Data Science Unconstrained Multivariate Optimization , Gradient (Steepest) Descent (OR) Learning Rule, Multivariate Optimization With Equality Constraints, Multivariate Optimization With Inequality Constraints, Introduction to Data Science, Solving Data Analysis Problems - A Guided Thought Process Predictive Modelling Linear Regression, Model Assessment, Diagnostics to Improve Linear Model Fit, Simple Linear Regression Model Building, Simple Linear Regression Model Assessment, Multiple Linear Regression, Cross Validation, Multiple Linear Regression Modelling Building and Selection, Classification, Logistic Regression, Performance Measures, Logistic Regression Implementation in R Algorithms K - Nearest Neighbors (kNN), K - Nearest Neighbors implementation in R, K - means Clustering, K - means Implementation in R				
Laboratory Work: NA				
Course Learning Objectives (CLO) The students will be able to: 1. Apply statistical methods and probability theory to analyze and interpret data using R language. 2. Create and interpret various types of data visualizations to communicate insights effectively. 3. Understand and apply basic machine learning algorithms to engineering problems. 4. Apply data science techniques to solve real-world engineering problems, such as predictive maintenance, optimization, and system modeling.				

Text Books

1. *APPLIED STATISTICS AND PROBABILITY FOR ENGINEERS – BY DOUGLAS MONTGOMERY*

Reference Books

1. *INTRODUCTION TO LINEAR ALGEBRA - BY GILBERT STRANG*